

Cloud Computing: Unit 3 Containerization and Kubernetes

VTU 2022 Scheme CBCS - BCS601 Module 3

--- MODULE OVERVIEW & LEARNING OBJECTIVES ---

1. Introduction to Linux Containers: cgroups, namespaces, and union file systems
2. Docker Architecture: Docker Daemon, CLI, Images, Containers, Registries
3. Kubernetes Architecture: Master Node [API Server, etcd, Scheduler] and Worker Nodes
4. Kubernetes Objects: Pods, ReplicaSets, Deployments, and Services [ClusterIP/NodePort]
5. Microservices Deployment and Container Orchestration on AWS/GCP

--- IMPORTANT VTU 10-MARK QUESTIONS ---

- Q1. Draw and explain Kubernetes cluster architecture [10 Marks]
- Q2. Differentiate Virtual Machines vs Docker Containers in detail [10 Marks]